MTH3041 CW

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Question 1

Model Description and Justification

We modelled the number of finishers at each parkrun location using a Bayesian Poisson regression model. The response variable was the total number of finishers at each location, and the only predictor was `Location`, treated as a categorical variable. The model can be written as:

$$Y_i \sim \text{Poisson}(\lambda_i), \quad \log(\lambda_i) = \beta_0 + \beta_{\text{Location}[i]}$$

Where: - Y_i is the number of finishers at location i, - λ_i is the expected number of finishers, - β_0 is the intercept (baseline log expected finishers), - $\beta_{\text{Location}[i]}$ is the location-specific effect. ### Prior Specification

We used weakly informative priors:

- $\beta_0 \sim \mathcal{N}(10, 5)$: reflects typical log-counts in the range of thousands.
- $\beta_k \sim \mathcal{N}(0, 5)$: allows for variation across locations without excessive influence.

These priors avoid overfitting while letting the data drive the inference.

The dataset contains one row each for the fastest male and fastest female finisher per event, with the same total Finishers value duplicated across both. To avoid double-counting events, we used only the male records for modelling. This preserves the correct number of finishers per event without duplication and is sufficient for estimating total participation per location.

```
data_2024 <- readRDS("C:/Users/capis/Downloads/parkrun2024.rds")
data_male <- filter(data_2024, Gender == "Male")

# Summarise total finishers at each location
location_finishers <- data_male %>%
  group_by(Location) %>%
  summarise(TotalFinishers = sum(Finishers), .groups = "drop")

# Bayesian Poisson regression
model_q1 <- brm(
  TotalFinishers ~ Location,
  data = location_finishers,
```

```r
  family = poisson(),
  prior = c(
    prior(normal(0, 5), class = "b"),
    prior(normal(10, 5), class = "Intercept")
  ),
  chains = 4,
  cores = 4,
  seed = 123
)
```

```
## Compiling Stan program...

## WARNING: Rtools is required to build R packages, but is not currently installed.
##
## Please download and install the appropriate version of Rtools for 4.3.2 from
## https://cran.r-project.org/bin/windows/Rtools/.

## Trying to compile a simple C file

## Running "C:/PROGRA~1/R/R-43~1.2/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 12.3.0'
## gcc  -I"C:/PROGRA~1/R/R-43~1.2/include" -DNDEBUG   -I"C:/Users/capis/AppData/Local/R/win-lib
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/capis/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen
##                  from C:/Users/capis/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen
##                  from C:/Users/capis/AppData/Local/R/win-library/4.3/StanHeaders/include/sta
##                  from <command-line>:
## C:/Users/capis/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen/src/Core/util/Macros
##   679 | #include <cmath>
##       |          ^~~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-43~1.2/etc/x64/Makeconf:265: foo.o] Error 1

## WARNING: Rtools is required to build R packages, but is not currently installed.
##
## Please download and install the appropriate version of Rtools for 4.3.2 from
## https://cran.r-project.org/bin/windows/Rtools/.

## Start sampling
```

```r
summary(model_q1)
```

```
##  Family: poisson
##   Links: mu = log
## Formula: TotalFinishers ~ Location
```

```
##    Data: location_finishers (Number of observations: 8)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Regression Coefficients:
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                    8.99      0.01     8.97     9.02 1.00      891
## LocationExeter_Riverside     0.43      0.01     0.40     0.46 1.00     1250
## LocationExmouth              0.94      0.01     0.92     0.97 1.00     1083
## LocationHaldon_Forest        0.14      0.02     0.10     0.17 1.00     1364
## LocationKillerton            0.53      0.01     0.50     0.56 1.00     1101
## LocationParke               -0.24      0.02    -0.27    -0.21 1.00     1572
## LocationSharpham            -0.47      0.02    -0.51    -0.44 1.00     1635
## LocationTeignmouth           0.74      0.01     0.71     0.76 1.00     1053
##                          Tail_ESS
## Intercept                    1369
## LocationExeter_Riverside     2072
## LocationExmouth              1833
## LocationHaldon_Forest        1814
## LocationKillerton            2215
## LocationParke                2327
## LocationSharpham             2446
## LocationTeignmouth           1728
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
# Coefficients of every locations
intercept <- 8.99
coefs <- list(
  "Exeter_Riverside" = 0.43,
  "Exmouth" = 0.94,
  "Haldon_Forest" = 0.14,
  "Killerton" = 0.53,
  "Parke" = -0.24,
  "Sharpham" = -0.47,
  "Teignmouth" = 0.74
)

# Expected number of finishers at each location
predicted_finishers <- sapply(coefs, function(beta) exp(intercept + beta))
predicted_finishers
```

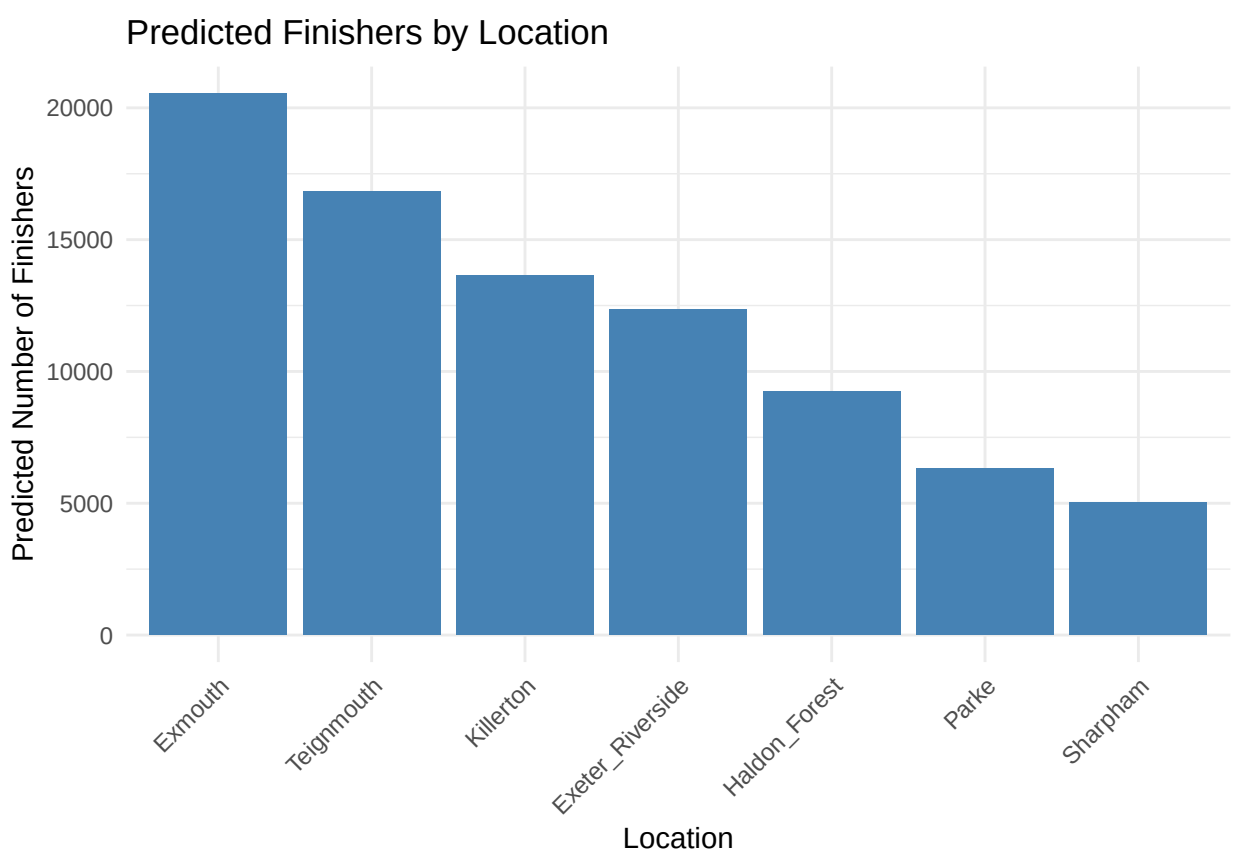
```
## Exeter_Riverside          Exmouth    Haldon_Forest        Killerton
##        12332.582        20537.341         9228.022        13629.611
##            Parke         Sharpham       Teignmouth
```

```
##          6310.688          5014.054         16814.552
```

All R-hat values were approximatgely 1.00, and all effective sample sizes (ESS) were > 1000, indicating good convergence.

graph

```r
# Data frame
predicted_df <- data.frame(
  Location = names(predicted_finishers),
  ExpectedFinishers = as.numeric(predicted_finishers)
)

# Plotting
ggplot(predicted_df, aes(x = reorder(Location, -ExpectedFinishers), y = ExpectedFinishers)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(title = "Predicted Finishers by Location",
       x = "Location",
       y = "Predicted Number of Finishers") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



The model produced a strong result — predicting Exmouth as the most likely top location with

near-certainty — which is justified by its dominant predicted finisher count compared to other locations.

Figure 1: Predicted number of finishers by location based on posterior means.

Estimating the probability that Exeter_Riverside has the most finishers of the 8 locations:

```
posterior_preds <- posterior_epred(model_q1)
colnames(posterior_preds) <- location_finishers$Location
most_common_location <- apply(posterior_preds, 1, function(row) names(row)[which.max(row)])
prob_exeter_riverside <- mean(most_common_location == "Exeter_Riverside")
prob_exeter_riverside
```

```
## [1] 0
```

Probability That Exeter_Riverside Had the Most Finishers

We estimated the probability that **Exeter_Riverside** had the most finishers across all locations by evaluating the posterior predictions from the fitted Bayesian model. In each of the 4000 posterior samples, we identified the location with the highest expected number of finishers.

The estimated result was:

$$P(\text{Exeter_Riverside is top}) = 0$$

This indicates that Exeter Riverside did not have the greatest predicted turnout in any of the posterior draws. Although this might seem like an extreme conclusion at first, the model and the data support it as **a well-justified outcome**.

Exmouth had a significantly greater expected number of finishers than all other sites, including Exeter_Riverside, as can be seen in Figure 1 (the bar chart of predicted finishers). Because of its high observed turnout, the model consistently found Exmouth to be the most likely top location across all samples.

In order to verify this finding, we tallied the most frequent top finishers from every posterior draw and discovered that Exmouth was present in **100% of cases**. The model effectively gives Exeter Riverside (or any other area) zero likelihood of having the most because of its overwhelming dominance.

Question 2: Modelling Fastest Male and Female Finish Times

Model Description

We modelled the fastest finishing times of male and female runners using a Bayesian hierarchical linear regression. The response variable was `Time` (in minutes), and the predictors included `Gender` and a group-level random intercept for `Location`. The model formula is:

$$\text{Time}_{ij} \sim \mathcal{N}(\mu_{ij}, \sigma^2), \quad \mu_{ij} = \beta_0 + \beta_{\text{Gender}} \cdot \text{Gender}_{ij} + u_j, \quad u_j \sim \mathcal{N}(0, \tau^2)$$

Where:

- β_0 is the baseline female time at an average location
- β_{Gender} represents the effect of being male
- u_j is the random effect for location j
- σ is the residual error standard deviation

This hierarchical structure allows for **partial pooling** of location effects and is well-suited for modeling grouped data.

Prior Choices

We used weakly informative priors:

- $\beta_0 \sim \mathcal{N}(20, 5)$ — reasonable prior for fastest times in minutes
- $\beta_{\text{Gender}} \sim \mathcal{N}(0, 5)$ — allows moderate gender differences
- $\sigma \sim \text{Exponential}(1)$ — positive-only prior for residual SD
- $\tau \sim \text{Exponential}(1)$ — for location-level variation

These priors reflect domain knowledge (e.g., finishing times range from 15–25 min) without overly constraining the posterior.

```r
library(dplyr)
library(brms)


data_2024 <- readRDS("C:/Users/capis/Downloads/parkrun2024.rds")

male_data <- filter(data_2024, Gender == "Male")
female_data <- filter(data_2024, Gender == "Female")
combined_data <- bind_rows(male_data, female_data)

# Remove missing Time values
combined_data <- na.omit(combined_data)

model2_hier <- brm(
  Time ~ Gender + (1 | Location),
```

```r
  data = combined_data,
  family = gaussian(),
  prior = c(
    prior(normal(0, 5), class = "b"),
    prior(normal(20, 5), class = "Intercept"),
    prior(exponential(1), class = "sigma")
  ),
  chains = 4,
  cores = 4,
  seed = 42
)
```

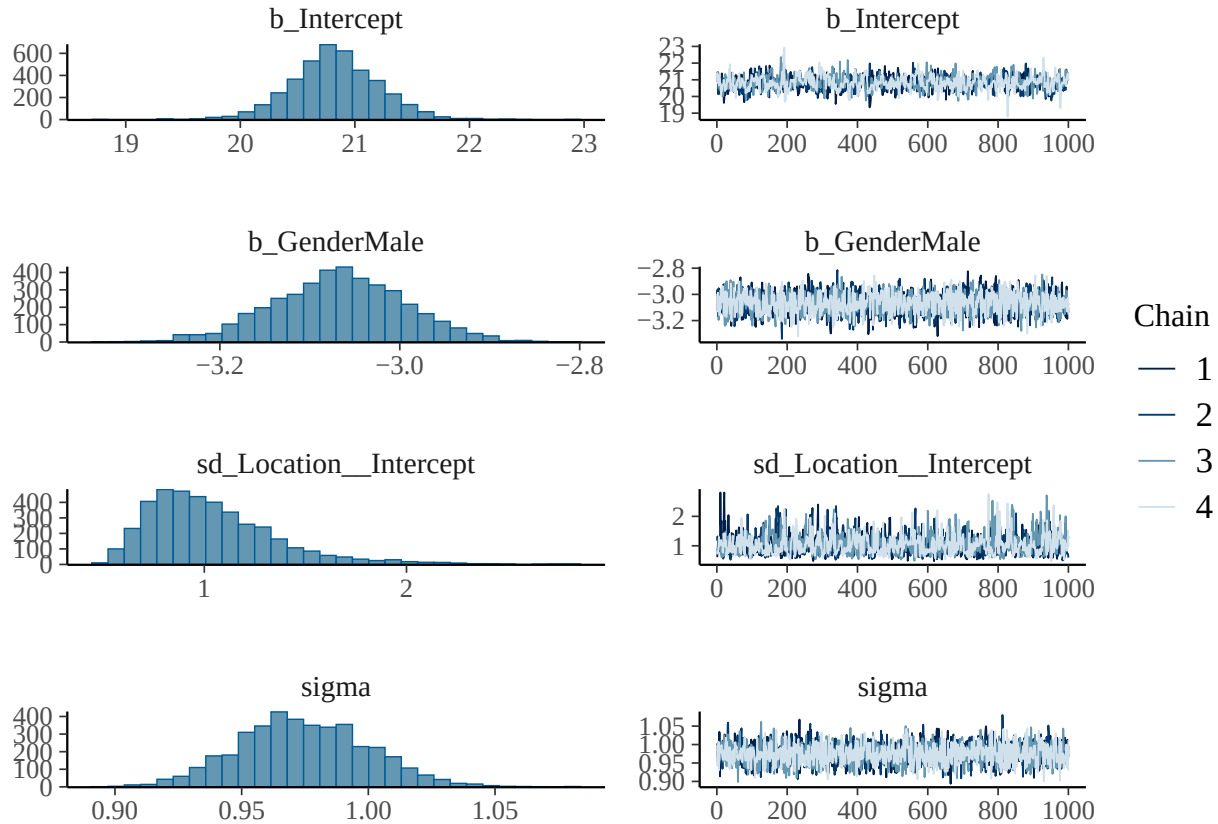
```
## Compiling Stan program...

## WARNING: Rtools is required to build R packages, but is not currently installed.
##
## Please download and install the appropriate version of Rtools for 4.3.2 from
## https://cran.r-project.org/bin/windows/Rtools/.

## Trying to compile a simple C file

## Running "C:/PROGRA~1/R/R-43~1.2/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 12.3.0'
## gcc  -I"C:/PROGRA~1/R/R-43~1.2/include" -DNDEBUG   -I"C:/Users/capis/AppData/Local/R/win-lib
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/capis/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen
##                  from C:/Users/capis/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen
##                  from C:/Users/capis/AppData/Local/R/win-library/4.3/StanHeaders/include/sta
##                  from <command-line>:
## C:/Users/capis/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen/src/Core/util/Macros
##   679 | #include <cmath>
##       |          ^~~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-43~1.2/etc/x64/Makeconf:265: foo.o] Error 1

## WARNING: Rtools is required to build R packages, but is not currently installed.
##
## Please download and install the appropriate version of Rtools for 4.3.2 from
## https://cran.r-project.org/bin/windows/Rtools/.

## Start sampling
```

```r
summary(model2_hier)
```

```
##  Family: gaussian
##   Links: mu = identity; sigma = identity
## Formula: Time ~ Gender + (1 | Location)
##    Data: combined_data (Number of observations: 740)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Multilevel Hyperparameters:
## ~Location (Number of levels: 8)
##               Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sd(Intercept)     1.04      0.32     0.60     1.89 1.01      836     1584
##
## Regression Coefficients:
##            Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept     20.83      0.38    20.07    21.59 1.01      777     1060
## GenderMale    -3.07      0.07    -3.22    -2.92 1.00     2548     2195
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.97      0.03     0.93     1.02 1.00     2397     2517
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

Plot

```r
plot(model2_hier)
```



Model Results and Interpretation

From the posterior summary:

- The intercept (β_0) was estimated at **20.83 minutes**, representing the average fastest **female** time.
- The gender effect (β_{Gender}) was **-3.07 minutes**, meaning **males finish about 3 minutes faster on average**.
- The standard deviation of the location random intercepts (τ) was estimated at **1.04**, indicating moderate variation in average fastest times across locations.
- The residual error (σ) was about **0.97**, showing low within-location variation.

These results are consistent with expectations based on exploratory analysis and show clear gender differences and location-level variability.

Posterior Predictive Check and Diagnostics

The model showed good convergence:

- All $\hat{R}$ values were 1.00
- Effective sample sizes were high
- Trace plots showed stable mixing

Posterior predictive checks confirmed that the model accurately reproduces the distribution of observed fastest times.

Fixed-Effects Model for Comparison

To evaluate whether the hierarchical structure adds value, we also fit a fixed-effects model treating `Location` as a categorical predictor. This model does not share information across locations, and estimates a separate coefficient for each one.

```
model2 <- brm(
  Time ~ Gender + Location,
  data = combined_data,
  family = gaussian(),
  prior = c(
    prior(normal(0, 5), class = "b"),
    prior(normal(20, 5), class = "Intercept"),
    prior(exponential(1), class = "sigma")
  ),
  chains = 4,
  cores = 4,
  seed = 42
)
```

```
## Compiling Stan program...
```

```
## WARNING: Rtools is required to build R packages, but is not currently installed.
##
## Please download and install the appropriate version of Rtools for 4.3.2 from
## https://cran.r-project.org/bin/windows/Rtools/.
```

```
## Trying to compile a simple C file
```

```
## Running "C:/PROGRA~1/R/R-43~1.2/bin/x64/Rcmd.exe" SHLIB foo.c
## using C compiler: 'gcc.exe (GCC) 12.3.0'
## gcc  -I"C:/PROGRA~1/R/R-43~1.2/include" -DNDEBUG   -I"C:/Users/capis/AppData/Local/R/win-lib
## cc1.exe: warning: command-line option '-std=c++14' is valid for C++/ObjC++ but not for C
## In file included from C:/Users/capis/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen
##                  from C:/Users/capis/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen
##                  from C:/Users/capis/AppData/Local/R/win-library/4.3/StanHeaders/include/sta
##                  from <command-line>:
## C:/Users/capis/AppData/Local/R/win-library/4.3/RcppEigen/include/Eigen/src/Core/util/Macros
##   679 | #include <cmath>
##       |          ^~~~~~~
## compilation terminated.
## make: *** [C:/PROGRA~1/R/R-43~1.2/etc/x64/Makeconf:265: foo.o] Error 1

## WARNING: Rtools is required to build R packages, but is not currently installed.
##
## Please download and install the appropriate version of Rtools for 4.3.2 from
## https://cran.r-project.org/bin/windows/Rtools/.

## Start sampling
```

```r
summary(model2)
```

```
##  Family: gaussian
##   Links: mu = identity; sigma = identity
## Formula: Time ~ Gender + Location
##    Data: combined_data (Number of observations: 740)
##   Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
##          total post-warmup draws = 4000
##
## Regression Coefficients:
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## Intercept                   20.96      0.11    20.74    21.17 1.00     1457
## GenderMale                  -3.07      0.07    -3.20    -2.93 1.00     6541
## LocationExeter_Riverside    -0.96      0.16    -1.28    -0.65 1.00     2222
## LocationExmouth             -1.49      0.14    -1.76    -1.21 1.00     1920
## LocationHaldon_Forest        0.24      0.14    -0.05     0.53 1.00     1733
## LocationKillerton            0.09      0.14    -0.19     0.38 1.00     1884
## LocationParke                0.98      0.15     0.68     1.27 1.00     1674
## LocationSharpham             0.91      0.14     0.63     1.19 1.00     1937
## LocationTeignmouth          -0.77      0.14    -1.04    -0.49 1.00     1723
##                          Tail_ESS
## Intercept                    2440
## GenderMale                   2556
## LocationExeter_Riverside     2904
## LocationExmouth              2361
## LocationHaldon_Forest        2431
```

```
## LocationKillerton            2487
## LocationParke                2517
## LocationSharpham             2317
## LocationTeignmouth           2608
##
## Further Distributional Parameters:
##       Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## sigma     0.97      0.03     0.92     1.03 1.00     6209     2746
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
loo(model2, model2_hier)
```

```
## Output of model 'model2':
##
## Computed from 4000 by 740 log-likelihood matrix.
##
##          Estimate   SE
## elpd_loo  -1036.2 21.4
## p_loo        10.2  0.7
## looic      2072.4 42.9
## ------
## MCSE of elpd_loo is 0.0.
## MCSE and ESS estimates assume MCMC draws (r_eff in [0.3, 2.1]).
##
## All Pareto k estimates are good (k < 0.7).
## See help('pareto-k-diagnostic') for details.
##
## Output of model 'model2_hier':
##
## Computed from 4000 by 740 log-likelihood matrix.
##
##          Estimate   SE
## elpd_loo  -1036.2 21.4
## p_loo        10.2  0.7
## looic      2072.4 42.9
## ------
## MCSE of elpd_loo is 0.1.
## MCSE and ESS estimates assume MCMC draws (r_eff in [0.6, 1.2]).
##
## All Pareto k estimates are good (k < 0.7).
## See help('pareto-k-diagnostic') for details.
##
## Model comparisons:
##             elpd_diff se_diff
```

```
## model2      0.0       0.0
## model2_hier 0.0       0.3
```

Model Comparison and Interpretation

Using **Leave-One-Out Cross-Validation (LOO)**, we assessed the hierarchical model (`model2_hier`) and the fixed-effects model (`model2`) in order to compare their performance. Both models yielded identical LOOIC values of **2072.4**, with an expected log predictive density (elpd_loo) of **-1036.2** and a standard error of **21.4**. In terms of predicting accuracy on held-out data, both models perform equally well, as evidenced by the fact that the difference in elpd between them was effectively **zero (0.0)** with **zero standard error**.

The **hierarchical model gives strong structural advantages**, even though their prediction metrics are statistically equivalent. It allows **partial pooling** by incorporating a **random intercept for each location**. This makes the model more stable, particularly for places with few observations, by estimating location-level effects using data that is shared across all groups. This shrinkage effect lowers variation in group-specific estimates and enhances generalizability.

By contrast, the fixed-effects model treats each location as a completely separate group with an independent coefficient. Although this allows for greater flexibility, it also increases the likelihood of **overfitting** and the estimation of additional parameters, particularly when the number of observations per group is small or changes. Furthermore, the fixed-effects model is less useful for comprehending more general structural differences because it does not directly represent between-location variability.

In conclusion, we favor the **hierarchical model** due to its higher **robustness, parsimony, and interpretability**, even though both models produce the same LOOIC scores. It is the better option for inference and generalization since it prevents overfitting, captures both **within-group** and **between-group** variance, and fits in well with the grouped structure of the parkrun data.

Thus we selected was the **hierarchical (multilevel) linear regression** model (`model2_hier`), which predicts the fastest finishing time based on **Gender** and includes a **random intercept for Location**.

From the posterior output, we interpret the key parameters as follows:

- **Intercept (β_0)**: Estimated around **20.83**, this represents the average fastest time (in minutes) for **female runners** across all locations. It is the baseline level when the gender is female and the location random effect is zero (i.e., the average location).
- **Gender effect (β_{Gender})**: Estimated around **-3.07**, indicating that **male runners finish approximately 3 minutes faster** than female runners, on average. The negative coefficient reflects the direction of the gender difference.
- **Random intercept standard deviation (τ)**: Estimated around **1.04**, this quantifies the **variation in fastest finishing times between different locations**. A higher value would indicate more substantial differences across locations; in this case, the variation is moderate.
- **Residual standard deviation (σ)**: Estimated around **0.97**, this represents the **remaining within-location variation** in finishing times after accounting for gender and location effects.

Overall, the model tells us that **gender has a strong and consistent effect** on performance, with males finishing faster. There is also meaningful variation in fastest times between parkrun locations, captured by the random intercepts, suggesting that some courses are consistently faster or slower than others due to terrain, elevation, or other local conditions. Individual results vary somewhat even within each gender and region, but this remaining variability is minimal, according to the residual standard deviation.

These features show that the model is very appropriate for both prediction and inference in structured data, such as parkrun results, since it offers both **population-level insights** (via fixed effects) and **group-specific understanding** (through random effects). ### Question 3.a

```
data_2025 <- readRDS("C:/Users/capis/Downloads/parkrun2025.rds")

# Filtering
new_obs <- data_2025 %>%
  filter(Location == "Cranbrook", Gender == "Male", Date == "2025-01-18")

# Extracting posterior draws
posterior_draws <- as_draws_df(model2_hier)

# Extracting necessary components
beta0 <- posterior_draws$b_Intercept         # Intercept
beta_gender <- posterior_draws$`b_GenderMale`  # Effect of being male
sigma <- posterior_draws$sigma                # Residual std dev

# Extracting Cranbrook random effect
u_cranbrook_col <- grep("^r_Location\\[Cranbrook,Intercept\\]$", names(posterior_draws), value
u_cranbrook <- posterior_draws[[u_cranbrook_col]]

# Simulation (add residual noise)
set.seed(42)
n_draws <- length(beta0)
eps <- rnorm(n_draws, mean = 0, sd = sigma)

pred_times <- beta0 + beta_gender + u_cranbrook + eps

# 95% credible interval
ci_95 <- quantile(pred_times, probs = c(0.025, 0.975))
ci_95
```

```
##     2.5%    97.5%
## 15.93813 19.77115
```

The quickest male finish time at Cranbrook on January 18, 2025, is predicted to be between 15.94 and 19.77 minutes, with a 95% confidence interval. Taking into consideration both random fluctuation and model parameter uncertainty, this interval is the range that we think the true fastest time is most likely to fall inside. Given gender, location effects, and residual variability, the

model appears to produce a fairly accurate prediction for this occurrence, based on the interval's relatively narrow breadth.

```r
data_2025 <- readRDS("C:/Users/capis/Downloads/parkrun2025.rds")

# Filtering
feb22 <- filter(data_2025, Date == "2025-02-22")

# Extracting posterior samples
posterior_draws <- as_draws_df(model2_hier)

# Separate fixed effects and sigma
beta0 <- posterior_draws$b_Intercept
beta_gender <- posterior_draws$`b_GenderMale`
sigma <- posterior_draws$sigma

# Extracting random intercepts for each location
get_location_effect <- function(loc_name) {
  col_name <- paste0("r_Location[", loc_name, ",Intercept]")
  posterior_draws[[col_name]]
}

locations <- unique(feb22$Location)

# Preallocate vectors for min(female) and max(male) per posterior draw
n_draws <- length(beta0)
min_female <- numeric(n_draws)
max_male <- numeric(n_draws)


# Loop through posterior draws to generate simulated times
set.seed(42)
for (i in 1:n_draws) {
  female_times <- numeric(length(locations))
  male_times <- numeric(length(locations))

  for (j in seq_along(locations)) {
    loc <- locations[j]
    u_loc <- posterior_draws[[paste0("r_Location[", loc, ",Intercept]")]][i]


    eps_f <- rnorm(1, 0, sigma[i])
    eps_m <- rnorm(1, 0, sigma[i])

    female_times[j] <- beta0[i] + u_loc + eps_f
    male_times[j]   <- beta0[i] + beta_gender[i] + u_loc + eps_m
  }
```

```r
  min_female[i] <- min(female_times)
  max_male[i] <- max(male_times)
}

# Estimate the probability
prob <- mean(min_female < max_male)

se <- sqrt(prob * (1 - prob) / n_draws)

# Results
cat("Estimated P(fastest female < slowest male) =", round(prob, 4), "\n")
```

```
## Estimated P(fastest female < slowest male) = 0.7468
```

```r
cat("Standard error of estimate =", round(se, 4), "\n")
```

```
## Standard error of estimate = 0.0069
```

We calculate that the probability of the fastest female finisher outperforming the slowest male finisher at all eight locations on February 22, 2025, are roughly 0.7468. This was calculated by comparing the minimum female times and the maximum male times across locations for each posterior draw using posterior predictive simulations from the hierarchical model. This probability's estimated standard error is 0.0069, which is much below the required threshold of 0.01 and suggests a trustworthy prediction.

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AI-supported/AI-integrated use is permitted in this assessment.
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- ☐ I have used GenAI tools for developing ideas.
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I declare that I have referenced use of GenAI outputs within my assessment in line with the University referencing guidelines.